

Introduction to Web

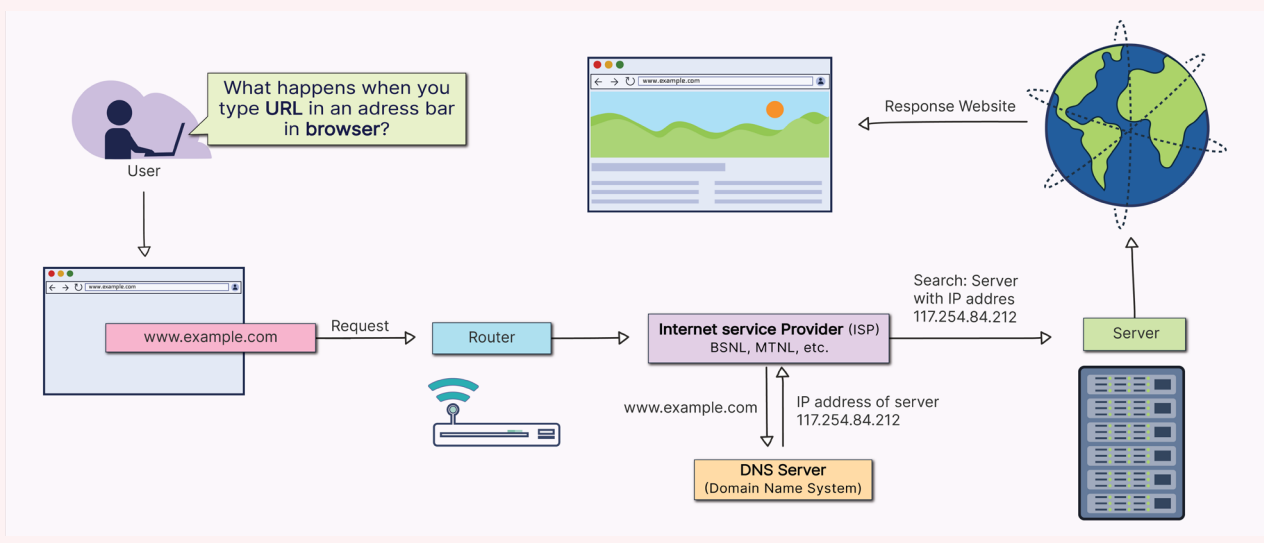


Do you know how a website loads when we type www.google.com in the browser? How does the internet work? Let's learn how!



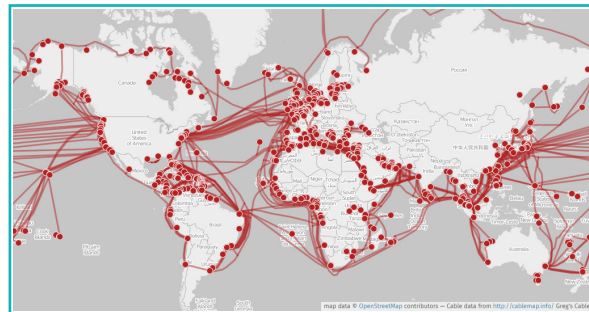
How does the internet work?

1. When we want to access a website over the internet, we type a website's domain name in the browser address window.
2. If connected to the internet, our computer or client sends an information request through the Internet Service Provider (ISP). ISP connects with the DNS server to get the IP address for a specific domain name.
3. With the help of an IP address and a router, ISP sends a request for the data to a specific server.
4. Once the request is received, the server checks if the data is available. It sends data back to the browser, which displays it in the form of a website.



The internet is an immensely large network of wires that connects digital devices and systems. It is used to transfer information from one location to another in the form of electrical signals. Digital devices like computers, smartphones etc. use the internet to share data and communicate with each other.

The internet connects the world primarily through a huge network of cables. These cables are connected deep underground. They provide all the internet connections throughout the country, including the ones at our homes.



Source: commons.wikimedia.org

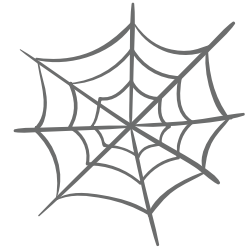
Applications of the internet

The internet has become an essential part of our lives. It is used for many purposes:

- To read news, books, articles from around the world
- To search for information about a topic
- To communicate with friends and relatives
- To share ideas, pictures, videos with others
- To watch movies and listen to music
- To play games, make payments and shop



Just like a spider web, different computers are connected to each other through the internet. Such a huge network of connected computers all over the world is called the **World Wide Web (WWW)**.



The **World Wide Web (www)**, commonly known as **the Web**, is a system where digital information is identified and accessed over the Internet.



We all know that computers have a memory where we store information. Computers can share all this information with each other through the web.

We can access information over the internet through a browser on our desktop computers, laptops or phones.

Browser

A computer application that displays content from internet.

For example: Google Chrome, Microsoft Edge



The information is received and displayed in a specific format called a website.

A Website is a way to arrange information in a specific format. Each website has a specific address, and the internet uses this address to reach that website.

Now, let's understand how we access a website and how the web works.

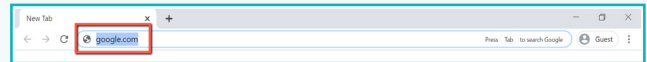
The web works like a mail or parcel delivery system. Whenever we send a mail or parcel to anyone, we need their address. Similarly, to access information on the internet we need the website's address.

Whenever we access a website, the computer sends information over the web. As a response, the information is received and displayed in the browser in the form of a website.

Activity

Understand the working of the internet

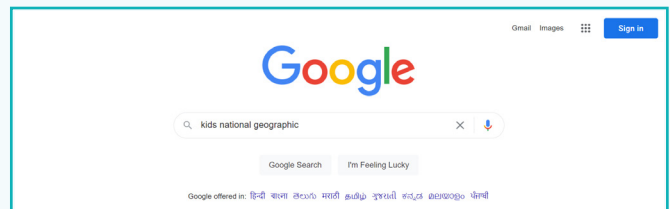
- 1 Open a browser on the computer
- 2 Type google.com in the browser and click enter



How do we find a specific website from pool of so many of websites. There are special websites called Search Engines which help us find other websites.

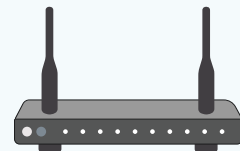
Search engines use keywords or domain names to find a specific website or webpage.

For example: kiddle.com, google.com



When we type the website address, a machine called a router finds it.

Router: A Router is a device that helps devices communicate with the web. It decides a route or path between devices at home or between clients and servers for data transfer.



This device acts as a sorting machine, where packages and mail are picked up and sorted to be delivered on a specific path according to their delivery address, like our manual delivery system.



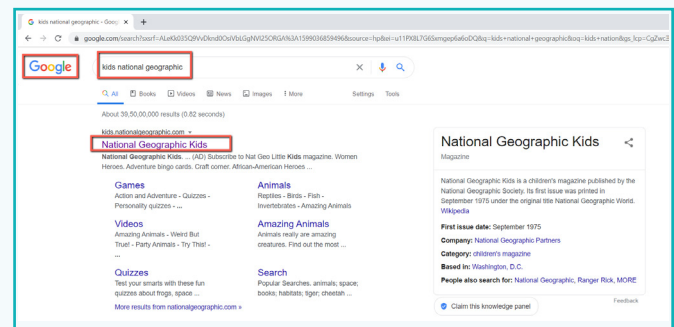
Bytes

Do you know how many active websites were available on the internet in 2020? Approximately over 200 million! That is 8 0s after 2 or 200000000!

Activity

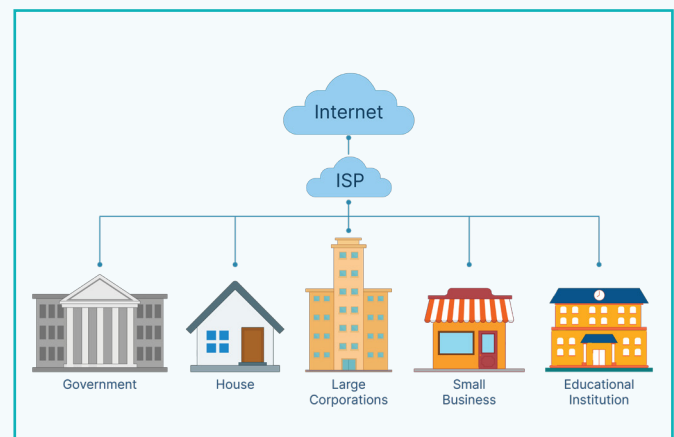
- 3 Search for “kids national geographic”. There will be multiple search results. Click on blue text (link) which says “National Geographic Kids”

Link: A link can be text or an image. When clicked, it takes the user to a new webpage. The webpage address is a part of the link which gathers content from the internet. This page address is also called a URL (Uniform Resource Locator).

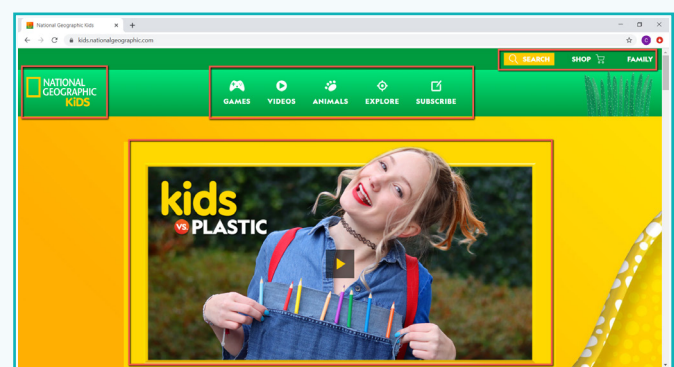


A router can only work if an internet connection is available. Internet is provided by different companies called Internet Service Providers (ISP). They are similar to courier services for mail and parcels.

For example – BSNL, MTNL etc.



Once we click on the link “kids.nationalgeographic.com”, a website opens.



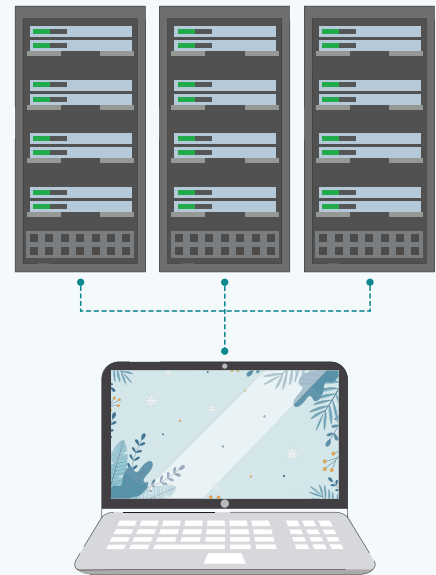
We have requested this website, which makes our computer a client. The special computer which shows us the website content becomes a server.

Activity

Server and Client

When we order a toy or an item, a request is sent to the shop which sells it. The shop finds the object in its warehouse and sends it to us.

When we enter the website address and press enter, a request is sent for data stored in a special computer. Similar to how objects are stored in a warehouse in an orderly manner, special computers that are always connected to the internet and store information or data are called **Servers**.



Devices which want to access information and receive it are called **Clients**.

For example: our computers, laptops, mobile phones are clients.

Exercise

Answer in one line

1 What is World Wide Web?



Whenever we place an order online, we have to provide our address for delivery. Similarly, to send or receive requests for information, we need the addresses for the client and server.

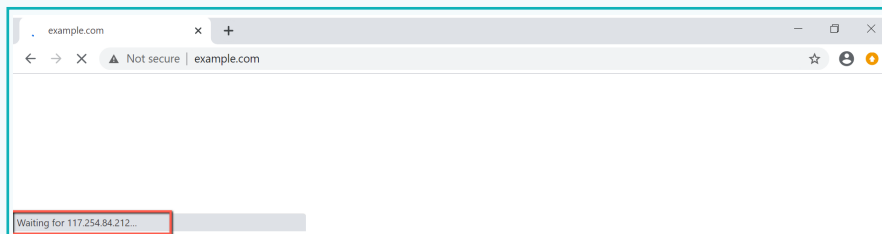
Activity

IP Address: All devices connected to the internet have an address called an IP (Internet Protocol) address. With the help of an IP address, devices can send information from specific servers to clients, or vice versa.

For example: An IP address is represented by a series of numbers that identify a specific computer or server on the internet. A typical IP address looks like this: 66.249.66.1



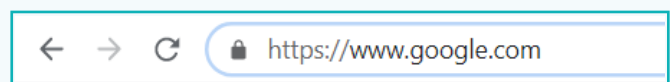
When we access the website “example.com”, we can observe its IP address in the bottom left corner of the browser.



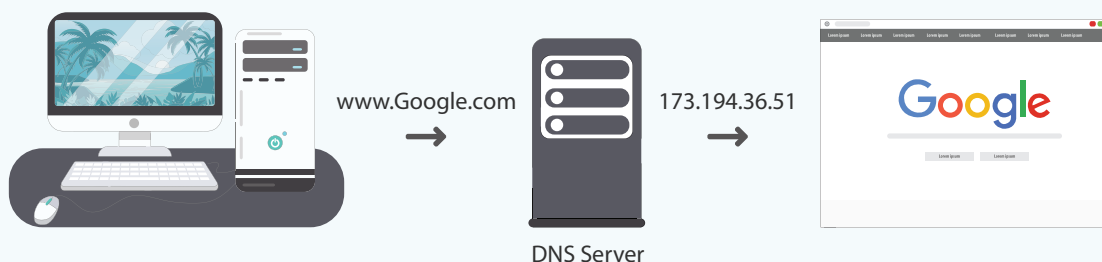
But will it be possible for us to remember a website address in the form of such numbers? It seems difficult. Instead, we can access websites using a unique address with a specific name. This is called a domain name. In this case, it was example.com.

Domain name - It is a website address that we type in the browser address bar to visit the website. Each website is identified using a specific domain name.

For example: To visit the Google website, we type the domain name <https://www.google.com>.



There are some Special servers that match a website’s domain name and IP address. It is similar to how roll numbers are linked to names in a class.



For each domain name, a specific IP address of a server is registered. The servers which store an IP address for each domain name are called **Domain Name Systems (DNS)** servers.

Exercise

Fill in the blanks

- Special computers that store data and send it over the internet when requested by clients are called _____.
- Servers which store the destination web server address are called _____ servers.

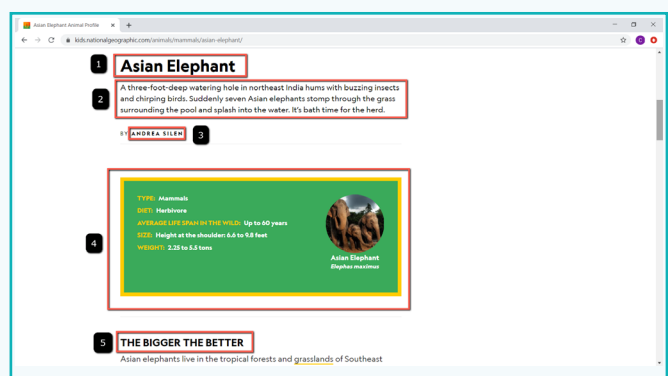
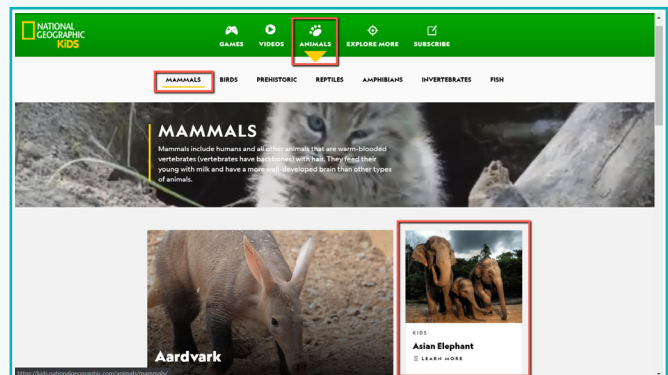
Activity

A web page is content displayed in web browser as a part of a website. It can have different resources to share information. For example: text with different font colours, information about the page style which controls the look and layout, and media such as images, sounds, and videos.

- From options, click on the 'animals' link and click on the 'mammals' option

Look for an image of Asian elephants and click on the **image link**.

- A webpage with information on Asian Elephants will appear. Observe the different components like the title, sub-title, paragraph, images etc.



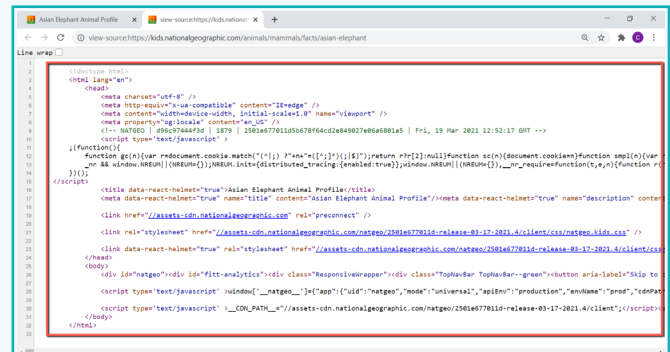
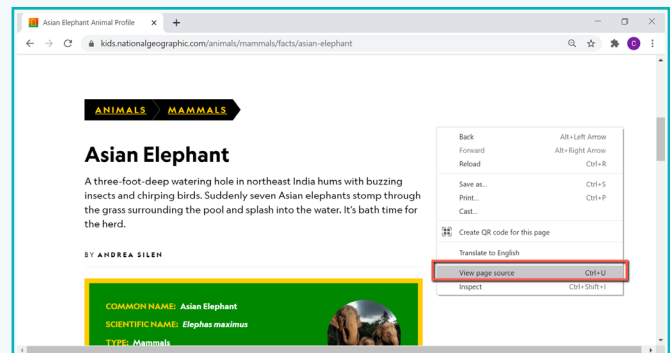
Activity

Right click on the page and select the option “View page source”.

A new window will appear with some text on it. This is the language a web browser understands. Based on this, it displays the webpage that we see.

We have seen how the internet works and the various elements that are necessary for its working. We also learnt about search engines, websites and the webpage.

In the next lesson, we will learn how to create a webpage by learning this language and its rules that a web browser understands.



Exercise

Choose the correct description of the marked sections in the following webpage based on the section numbers

1

Asian Elephant

2

A three-foot-deep watering hole in northeast India hums with buzzing insects and chirping birds. Suddenly seven Asian elephants stomp through the grass surrounding the pool and splash into the water. It's bath time for the herd.

BY ANDREA SILEN

3

4

TYPE: Mammals

DIET: Herbivore

AVERAGE LIFE SPAN IN THE WILD: Up to 60 years

SIZE: Height at the shoulder: 6.6 to 9.8 feet

WEIGHT: 2.25 to 5.5 tons



Asian Elephant
Elephas maximus

5

THE BIGGER THE BETTER

Asian elephants live in the tropical forests and grasslands of Southeast

- 4 Which section contains the image?
 - a. 1 ☐
 - b. 3 ☐
 - c. 4 ☐
 - d. 5 ☐
- 5 Which section is the heading?
 - a. 2 ☐
 - b. 3 ☐
 - c. 4 ☐
 - d. 1 ☐
- 6 Which section is the text paragraph?
 - a. 1 ☐
 - b. 2 ☐
 - c. 3 ☐
 - d. 5 ☐
- 7 Which section is the subtitle?
 - a. 5 ☐
 - b. 3 ☐
 - c. 2 ☐
 - d. 1 ☐

The internet is a large network connecting computers, servers, and devices.



Tech Flash

- **World Wide Web (www)** : the Web is system where data can be accessed over the Internet.
- **Browser** : a software that displays content over the web in the form of websites.
- **Website** : is set of information expressed in a specific format that can be accessed over the internet with the help of a browser.
- **Router** : a device that helps other devices choose a path and transfer data over the internet.
- **ISP** : Internet Service Provider is a company that provides an internet service.
- **Servers** : are special computers which connect to the web and store data.
- **Clients** : devices who access information from the web.
- **IP Address** : Address is a connected device's unique address which is used to send or receive data over the web.
- **Domain name** : a website name that we type in the browser to visit the website.
- **DNS Server** : special servers that store web server addresses where a website is stored.
- **Search Engines** : special websites which use keywords to search for other websites.
- **Link** : text or an image on a webpage. When clicked, it takes us to a new webpage or to a specific part of the webpage.
- **Web page** : a document on a website which has its own unique address.